Md Messal Monem Miah

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LinkedIn | GitHub | Google Scholar

Research Interests

Large Language Models | Multimodality | Dialog Modeling | Information Retrieval

Education

Texas A&M University — College Station, Texas
Ph.D. in Computer Science; Supervisor: Dr. Ruihong Huang — Aug 2021–Present
CGPA: 4.0/4.0

Bangladesh University of Engineering and Technology — Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering — Feb 2015–Apr 2019

Work Experience

Texas A&M University — College Station, Texas
Research Assistant, Natural Language Processing Lab — Aug 2021 –Present

Google DeepMind — Mountain View, California
Research Intern — May 2025–Aug 2025
Project: Extending Tool Use Capabilities of LLM Agents from Native Audio Instructions.

Amazon — Boston, Massachusetts
Applied Scientist Intern — May 2024–Aug 2024
Project: Coherent Multimodal Answer Generation for AI Agents.

Infinitus Systems, Inc. — San Francisco, California
NLP/Speech Research Intern — Aug 2023–Dec 2023
Project: Multimodal Dialog Breakdown Detection for Automated Dialog Agents.

Amazon Lab 126 — Sunnyvale, California
Applied Scientist Intern — May 2022–Aug 2022
Project: Mitigating HVAC Noise in Ultrasound Based Respiration Rate Monitoring System.

Research Experience

Multimodal Retrieval for Contextual Long-form Evaluation — Oct 2023–Present
Developed a novel benchmark for multimodal retrieval with interleaved images and text, designed to challenge SOTA retrievers with multiple images and longer context lengths. Evaluated various unimodal and multimodal retrievers like BM25, Contriever, ColBERT, CLIP, BLIP, FLMR in zero and fewshot setups. Additionally, developed the MultiFLMR model to improve retrieval efficiency on the proposed benchmark.

Coherent Multimodal Answer Generation for AI Agents — May 2024–Aug 2024
Created a dataset focused on generating factually consistent and coherent multimodal answers, combining text and infographic images. Addressed the challenge of separately retrieved/generated text and images by ensuring alignment between the two in terms of optimal placement and citation of the image within the response. Evaluated the capabilities of commercial and open-source LLMs (Claude3, LLaMA3.1, Mistral, Mixtral) using various prompting techniques, including zero-shot, few-shot, self-consistency, and ReACT.

Multimodal Dialog Breakdown Detection for Automated Dialog Agents — Aug 2023–Nov 2023
Designed an end-to-end dialog breakdown detection system for a healthcare-focused automated phone-call agent. Leveraged LLMs like RoBERTa and acoustic models such as Wav2Vec2 to identify breakdowns caused by upstream ASR errors or intent misclassification. Achieved ∼70% accuracy, enabling timely human intervention to maintain call quality. The system outperformed Google's Gemini by 20% using in-context learning with up to 30 example calls.

Dialog Act (DA) Classification for Artificially Intelligent teammates Aug 2021–Aug 2023
Developed an online multimodal DA classification framework to enhance machine understanding of natural conversations. Leveraged raw audio and ASR transcripts of current and past utterances, integrating textual and acoustic features at a granular level. Proposed a novel Whisper based acoustic feature extraction method that helps achieve SOTA performance with a 3% increase in F1 score over existing models on MRDA & EMOTyDA datasets.

Publications

1. **Md Messal Monem Miah**, Adrita Anika, Xi Shi, Ruihong Huang, "**Hidden in Plain Sight: Evaluation of the Deception Detection Capabilities of LLMs in Multimodal Settings**," In Proceedings of the 2025 Conference of the Association for Computational Linguistics (ACL).
2. **Md Messal Monem Miah**, Agent Chatterjee, Arindam Mitra, Ruihong Huang, Man Luo, "**MIRACLE: Multimodal Image-text Retrieval and Analysis for Contextual Long-form Evaluation**," Multimodal Arithmetic Reasoning Workshop, CVPR 2024
3. Adarsh Pyarelal, John M Culnan, Ayesha Qamar, Meghavarshini Krishnaswamy, Yuwei Wang, Cheonkam Jeong, Chen Chen, **Md Messal Monem Miah**, Shahriar Hormozi, Jonathan Tong, Ruihong Huang, "**MultiCAT: Multimodal Communication Annotations for Teams**," Findings of the Association for Computational Linguistics: **NAACL** 2025.
4. **Md Messal Monem Miah**, Ulie Schnaithmann, Arushi Raghuvanshi, Youngseo Son, "**Multimodal Contextual Dialogue Breakdown Detection for Conversational AI Models**" In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): Industry Track
5. Adrita Anika, **Md Messal Monem Miah**, Man Luo, "**Leveraging LLM-Generated QA Pairs for Biomedical Open-Domain Question Answering**," Workshop on Large Language Models and Generative AI for Health, AAAI 2025
6. Yuanyuan Lei, **Md Messal Monem Miah**, Ayesha Qamar, Sai Ramana Reddy, Jonathan Tong, Haotian Xu, Ruihong Huang, "**EMONA: Event-level Moral Opinions in News Articles**" In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL).
7. **Md Messal Monem Miah**, Adarsh Pyarelal, and Ruihong Huang, "**Hierarchical Fusion for Online Multimodal Dialog Act Classification**," In Findings of the 2023 Association for Computational Linguistics (**EMNLP**), Singapore.
8. **M. M. M. Miah**, R. J. Tazim, F. T. Johora, M. I. Al Imran, S. S. Surma, F. Islam, R. Shabab, C. Shahnaz and A. Subhana, "**Non-Invasive Bilirubin Level Quantification and Jaundice Detection by Sclera Image Processing**," 2019 IEEE Global Humanitarian Technology Conference (GHTC), Seattle, WA, USA
9. R. J. Tazim, **M. M. M. Miah**, S. S. Surma, M. T. Islam, C. Shahnaz and S. A. Fattah, "**Biometric Authentication Using CNN Features of Dorsal Vein Pattern Extracted from NIR Image**," TENCON 2018 - 2018 IEEE Region 10 Conference, Jeju, Korea (South)

Significant Projects

- Exploring Zero-Shot Retrieval Capabilities of Modern VLMs
- Multi Evidence Natural Language Inference on Clinical Trial Records for Breast Cancer
- Estimation of Public Speaking Anxiety from Bio-behavioral Data
- Application of Vein Pattern Based Biometric Authentication in Banking System

Skills

- **Programming:** Python, MATLAB, R, C/C++
- **Machine Learning:** PyTorch, Scikit-learn, Keras, Tensorflow, Hugging Face

Relevant Graduate Course-works

Large Language Models | Vision Foundation Models | Deep Learning | Machine Learning | Data Mining & Analysis | Natural Language Processing | Information Storage & Retrieval | Analysis of Algorithms

Professional Services and Leadership Experience

- **Reviewer**, ACL 2025, EMNLP 2024, ACL 2024, EMNLP 2023, ACL 2023 System Demonstration
- **Treasurer**, Bangladesh Student Association, Texas A&M University

Awards and Honors

- **Dean's List Scholarship** (Departmental scholarship for academic excellence at all 4 undergraduate academic years)
- **1st Runners-up**, IEEE PES BDC DataThon, 2020 (A data science competition hosted in Kaggle)
- **Champion**, Inter University MATLAB Competition, Hacklab 1.0 (Organized by IEEE UIU Student Branch)
- **Champion**, Project Showcasing, Mecceleration 2018 (Organized by department of MCE, IUT)
- **Bronze Medal**, Techfest 2017 (Category – Digitize, Organized by IIT Mumbai)